

1. Applications Practice: Related Rates & Optimization

This set focuses on the applications you found challenging. Each problem includes strategy hints — use them if you get stuck.

1.1. Problem 1: Shadow Length (Related Rates)

A 2-meter tall man walks away from a 6-meter tall street lamp at a speed of 1.5 m/s. How fast is the length of his shadow increasing when he is 10 meters from the lamp?

Strategy hint: Draw the diagram. You'll have two similar right triangles. The lamp, the man, and the ground form one; the man, his shadow, and the ground form another.

1.2. Problem 2: Draining Tank (Related Rates)

A conical tank with height 12 m and radius 4 m at the top is draining at 2 m³/min. How fast is the water level falling when the water is 6 m deep?

Key relationship: For a cone, $V = \frac{1}{3}\pi r^2 h$. But r and h are related by similar triangles: $\frac{r}{h} = \frac{4}{12} = \frac{1}{3}$, so $r = \frac{h}{3}$. Substitute to get V in terms of h only.

1.3. Problem 3: Maximum Volume Box (Optimization)

A rectangular sheet of cardboard measures 20 cm by 30 cm. Equal squares are cut from each corner, and the sides are folded up to form an open-top box. What size squares should be cut to maximize the volume?

Strategy hint: If the square cut from each corner has side x , what are the dimensions of the box? What's the constraint on x ?

1.4. Problem 4: Closest Point on a Curve (Optimization)

Find the point on the curve $y = \sqrt{x}$ that is closest to the point $(4, 0)$.

Strategy hint: Minimize the **distance squared** (it's easier than minimizing distance, and gives the same answer). Distance squared from $(x, \sqrt{x})$ to $(4, 0)$ is $D^2 = (x - 4)^2 + (\sqrt{x} - 0)^2$.

1.5. Problem 5: Limit with L'Hôpital's Rule

Evaluate: $\lim_{x \rightarrow 0^+} x \ln x$

Strategy hint: This is an indeterminate form $0 \cdot (-\infty)$. Rewrite as a fraction to apply L'Hôpital's rule. Which way should you rewrite: $\frac{\ln x}{\frac{1}{x}}$ or $\frac{x}{\frac{1}{\ln x}}$? Try both mentally — which looks easier to differentiate?

1.6. Solutions

Solution 1: Let x = distance from man to lamp, s = shadow length. By similar triangles: $\frac{6}{x+s} = \frac{2}{s}$. Solve for relationship, differentiate with respect to t , substitute $\frac{dx}{dt} = 1.5$ and $x = 10$. Shadow grows at 0.75 m/s (independent of x !).

Solution 2: Substitute $r = \frac{h}{3}$ into volume: $V = \frac{\pi}{27}h^3$. Differentiate: $\frac{dV}{dt} = \frac{\pi}{9}h^2\frac{dh}{dt}$. With $\frac{dV}{dt} = -2$ and $h = 6$: $\frac{dh}{dt} = -\frac{1}{2\pi} \approx -0.159$ m/min.

Solution 3: Box dimensions: $(20 - 2x) \times (30 - 2x) \times x$. Volume $V(x) = 4x^3 - 100x^2 + 600x$. Constraint: $0 < x < 10$. Critical point: $V'(x) = 12x^2 - 200x + 600 = 0$. Solving: $x = \frac{25-5\sqrt{7}}{3} \approx 3.92$ cm (other root is outside domain).

Solution 4: $D^2 = (x - 4)^2 + x$. Derivative: $2(x - 4) + 1 = 0$, so $x = \frac{7}{2} = 3.5$. Closest point: $(3.5, \sqrt{3.5})$.

Solution 5: Rewrite: $\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}}$. This is $-\frac{\infty}{+}\infty$, so apply L'Hôpital: $\lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} (-x) = 0$.